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THE McGill PLANAR HYDROGEN ATMOSPHERE CODE (McPHAC)

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ABSTRACT

The McGill Planar Hydrogen Atmosphere Code (McPHAC) v1.1 calculates the hydrostatic equilibrium structure and emergent spectrum of an unmagnetized hydrogen atmosphere in the plane-parallel approximation, at surface gravities appropriate for neutron stars. McPHAC incorporates several improvements over previous codes for which tabulated model spectra are available: (1) Thomson scattering is treated anisotropically, which is shown to result in a 0.2%–3% correction in the emergent spectral flux across the 0.1–5 keV passband; (2) the McPHAC source code is made available to the community, allowing it to be scrutinized and modified by other researchers wishing to study or extend its capabilities; and (3) the numerical uncertainty resulting from the discrete and iterative solution is studied as a function of photon energy, indicating that McPHAC is capable of producing spectra with numerical uncertainties <0.01%. The accuracy of the spectra may at present be limited to ∼1%, but McPHAC enables researchers to study the impact of uncertain inputs and additional physical effects, thereby supporting future efforts to reduce those inaccuracies. Comparison of McPHAC results with spectra from one of the previous model atmosphere codes (NSA) shows agreement to $\lesssim$1% near the peaks of the emergent spectra. However, in the Wien tail a significant deficit of flux in the spectra of the previous model is revealed, determined to be due to the previous work not considering large enough optical depths at the highest photon frequencies. The deficit is most significant for spectra with $T_{\rm eff} < 10^{5.6}$ K, though even there it may not be of much practical importance for most observations.

Key words: stars: atmospheres – stars: neutron – X-rays: binaries – X-rays: general – X-rays: stars

1. INTRODUCTION

In recent years, there has been a large and growing interest in the application of calculated neutron star emergent spectral models to spectral observations in the optical and X-ray passbands. Reviews exist in two proceedings (Pavlov et al. 1995; Zavlin 2009), and various motivating examples are given below.

Following the suggestion that the emergent spectrum from a quiescent transient low-mass X-ray binary (qLMXB) should be expected to be an unmagnetized hydrogen (H) atmosphere spectrum (Brown et al. 1998), H atmosphere spectra have been used to measure the radiation radii from a large number of such systems. This includes ∼10 field qLMXBs, to which the distances are typically uncertain at the 25%–50% level (Rutledge et al. 1999, 2000, 2001a, 2001b, 2002b; Wijnands et al. 2003; Tomsick et al. 2004, 2005, 2007; Campana et al. 2004; Jonker et al. 2004, 2007a, 2007b; in't Zand 2005; Cackett et al. 2006, 2008; Cornelisse et al. 2007; Heinke et al. 2007; Degenaar et al. 2009). These modeling efforts place constraints on the measured radii[1] and have lent support to the interpretation of deep crustal heating in neutron stars (Haensel & Zdunik 1990, 2008; Brown et al. 1998) proposed to explain their luminosities. Also, H atmosphere spectral results from post-outburst transients (e.g., Cackett et al. 2008), as they cool (Ushomirsky & Rutledge 2001; Rutledge et al. 2002c), have had significant physical implications for the reactions that take place in the neutron star crust, including neutron star core cooling rates (Yakovlev et al. 2003; Shternin et al. 2007) and neutron superfluidity in the inner crust, combined with a very low impurity parameter (i.e., a very pure crust; Brown & Cumming 2009).

Unmagnetized hydrogen atmosphere models have been applied to study pulsars as well (Zavlin & Pavlov 1998; Zavlin et al. 1998, 2002; Zavlin 2006; Bogdanov et al. 2007), usually to derive limitations on the physical sizes of their emission areas.

Hydrogen atmosphere models have been used to derive neutron star mass–radius constraints from X-ray observations of qLMXBs in globular clusters, where the distance uncertainties are more typically ∼10% and lower, including in NGC 6440 (in't Zand et al. 2001), in Omega Cen (Rutledge et al. 2002a; Gendre et al. 2003b), in 47 Tuc (Heinke et al. 2003b, 2006), in NGC 6397 (Grindlay et al. 2001), in Terzan 1 (Wijnands et al. 2002), in M13 (Gendre et al. 2003a), in M28 (Becker et al. 2003), A-1 in M30 (Lugger et al. 2007), in M80 (Heinke et al. 2003a), in M55 and NGC 3201 (Webb et al. 2006), in NGC 2808 (Webb & Barret 2007), and in NGC 6304 (Guillot et al. 2009a, 2009b). These mass–radius constraints are interpreted to constrain the dense matter equation of state (EOS)—the relationship between pressure and density in nonrelativistic matter at and above nuclear density. As such, neutron star mass–radius constraints are used by nuclear physicists as guidance for examining the credibility of proposed EOSs (Lattimer & Prakash 2001, 2007). These constraints are therefore of strong observational interest in the nuclear physics community.

Thus, there are a large number of observational results, with important physics implications, that rest on the accuracy of the calculated hydrogen atmosphere models available for interpretation of observed spectra. Unmagnetized hydrogen atmosphere models are valid for $B \ll 10^{10}(T/10^6\ {\rm K})$ G (Zavlin et al. 1996) and are thus relevant to most of the examples given above.

Most applications of unmagnetized hydrogen atmosphere models in X-ray spectral fitting are through tabulated models that have been made available through XSPEC (Arnaud 1996). The main such models are NSA and NSAGRAV (Zavlin et al. 1996, hereafter Z96), HYD_SPECTRA (Gänsicke et al. 2002, hereafter

[1] What is typically measured is the radiation radius R_∞, which is related to the physical radius R through the neutron star mass M as $R_\infty = R/\sqrt{1-(2GM)/(Rc^2)}$. This measurement produces a joint mass–radius constraint.

G02), and NSATMOS (Heinke et al. 2006, hereafter H06). In addition, other unmagnetized hydrogen atmosphere models have previously been presented (e.g., Rajagopal & Romani 1996), as well as magnetic hydrogen atmosphere models that can be run in the zero field limit (e.g., Lloyd 2003), but these will not be considered in depth in the present work. NSA also allows for a nonzero magnetic field (Pavlov et al. 1995), as does NSMAX (Ho et al. 2008), which is another model implemented in XSPEC.

The three main unmagnetized hydrogen atmosphere models (Z96; G02; H06) have many similarities in the assumptions made and in the method of solution. Common to all three models are the assumptions of a plane-parallel atmosphere, isotropic and coherent unpolarized scattering, local thermodynamic equilibrium (LTE), radiative equilibrium, and hydrostatic equilibrium. Additional assumptions common to one or more models are complete ionization (H06), ideal EOS (H06), neglecting radiation pressure (Z96; G02), neglecting self-irradiation (Z96; G02), and neglecting electron conduction (Z96). The form of the radiative transfer equation solved in each of the three models is the same, including both a true absorption opacity and a scattering opacity. The calculation of the opacities, however, differs in the three models. In the first model (Z96), occupation probability formalism (Zavlin et al. 1994) is used to calculate the true absorption opacity, while the uncorrected Thomson scattering opacity is used. In the second model (G02), OPAL data (Iglesias & Rogers 1996) are used for the total (true absorption plus scattering) opacity, and the Thomson scattering opacity is corrected for collective effects (Boercker 1987). In the third model (H06), uncorrected thermal free–free true absorption opacity and uncorrected Thomson scattering opacity are used. All other differences between the three models relate to the way the coupled atmosphere structure and radiative transfer equations are solved, but these differences are not expected to lead to quantitative differences in the resulting spectra as long as adequate levels of discretization and iteration are used in each model.

The descriptions of the existing hydrogen atmosphere models (NSATMOS, NSA, NSAGRAV, and HYD_SPECTRA) that are present in the literature provide little guidance on the uncertainties in their resulting spectra. For example, NSA (Z96) states that the calculation was permitted to proceed until the fractional temperature correction at each depth was $\lesssim 1\%$. Because of their temperature correction procedure, this also constrains the variation of the total flux through the atmosphere, which is another common convergence criterion. However, constraining the variation of the total flux does not offer a reproducible constraint on the uncertainty in the emergent spectral intensity at a given photon energy; it may well be that the code implementation produces an underestimate of the intensity near the peak of the spectrum far in excess of 1% and a compensating overestimate of the intensity in the wings, or vice versa.

Thus, despite their wide use in astrophysics, the uncertainties on the available tabulated spectra are poorly quantified but are typically assumed to be $\lesssim 1\%$ in spectral intensity as a function of photon energy. To produce reliable mass–radius constraints from existing high-quality X-ray spectra, and even higher quality spectra that will be available from future X-ray missions, model H atmosphere spectra with quantified uncertainties in each spectral bin are required. In particular, simultaneous constraints on the mass and radius of a neutron star at the 5% level will require statistical uncertainties in each bin of the observed spectrum of better than 0.1%, so the model spectra should have uncertainties well below 0.1% for these to be negligible compared to the statistical uncertainty on the observed spectrum. Further, the calibration of the X-ray spectrometers will have to be improved accordingly, which is a tremendously difficult task for which there is no predictable timeline.

There are a number of factors that affect the uncertainty on computed emergent spectra, ranging from plain calculational uncertainty to the validity of the radiative transfer equations being solved and the accuracy of the input physics (e.g., opacities). All of these sources of uncertainty must be constrained to produce model spectra accurate to better than 0.1%.

The present work addresses the issue of spectral uncertainties in two ways. First, the numerical convergence of the current method is discussed at length in Section 4.3. Second, the source code is made available to the community, so that the uncertainties for specific parameter ranges, or due to the various inputs, can be probed by researches who wish to apply these models. The authors expect that numerical and model uncertainties will receive more attention in future code-presentation papers, and that other source codes will be made open source, such that more meaningful and extensive comparisons can be made between models/codes.

Open source code addresses an important issue in modern science, as has recently been discussed in the context of coming large projects (Weiner et al. 2009). Large private source codes cannot be scrutinized by the community, and the large complexity often deters other groups from independently reproducing the results. Thus, though the theoretical development is scrutinized in the peer-review process, the implementation is not, and coding bugs that significantly affect the results can go unnoticed for extended periods of time. One example of such a bug is the incorrect implementation of Fe opacity tables in a previous atmosphere model, uncovered after other codes produced discrepant results (see Pons et al. 2002).

The authors envisage the following as a possible path toward calculating more accurate unmagnetized hydrogen atmosphere emergent spectra. The present work releases the McGill Planar Hydrogen Atmosphere Code (McPHAC) to the community and demonstrates its capability to reach low computational uncertainty. Armed with this new tool, which allows the calculational uncertainties to be constrained, ourselves and others can then, for instance, study the sensitivity of the emergent spectra to uncertainties in the input opacities, identifying the parameter ranges for which it is the most important to better constrain those opacities. Further, physical effects presently ignored can be added to McPHAC to investigate their impact on the emergent spectra, or other codes can be developed including such effects and the results of these compared in detail to those of McPHAC. This all hinges on constraining the calculational uncertainties to be much smaller than the differences that arise from modifications to the physics and opacities, so illustrating how to do this is a main focus of the present work.

The calculational uncertainties will depend on the particular parameters run and on which physics is included (which may or may not presently be implemented in McPHAC). Thus, it is not possible to constrain the calculational uncertainties in general, but the overall capabilities of the code can be presented, along with examples to show how one can constrain the various calculational uncertainties as required in a particular case. This is done in the present work, enabling investigation of the impact of the common assumption of isotropic Thomson scattering.

This paper and the associated open source code (McPHAC) present the analytic equations of radiative transfer and atmospheric structure, their discretization, and the complete

details of the calculation of an emergent intensity spectrum, dependent on the neutron star mass and radius, and the effective temperature. The organization of this paper is as follows. In Section 2, the basic atmospheric model is described. The method of solution of the equations of radiative transfer and stellar structure is described in Section 3. In Section 4, there is discussion of some of the results of the calculation, including its convergence as a function of the depth and angular discretization, the effect of treating Thomson scattering anisotropically, and a comparison of the resulting spectra with one of the widely used existing spectral models (Z96). In Section 5, a number of possible improvements to the code are discussed, which may be necessary to improve the accuracy of the emergent spectra beyond the present approximate value of order percent in spectral intensity as a function of photon energy. In Section 6, the improvements offered by the present work and main results are summarized. Appendix A describes the temperature correction procedure, and Appendix B gives some description of the implementation of McPHAC and the external routines used.

2. ATMOSPHERE MODEL

This section presents a brief review of the atmosphere model used in the McPHAC code.

Many of the assumptions adopted in previous hydrogen atmosphere models are also made in the present work:

1. magnetic fields are neglected, limiting the validity of the computed spectra to atmospheres where $B \ll 10^{10}(T/10^6\,\mathrm{K})\,\mathrm{G}$ (Z96);
2. coherent scattering is assumed, which could lead to inaccuracies in the spectra at energies greater than 5 keV (1% of the electron rest mass);
3. hydrostatic and radiative equilibrium;
4. no electron conduction;
5. self-irradiation is ignored, limiting accuracy for neutron stars with $R < (3/2)R_S$ (H06; R_S is the Schwarzschild radius);
6. the plane-parallel assumption is used, since the scale height of the atmosphere is much smaller than the radius of the neutron star;
7. radiation pressure is neglected, which is appropriate for atmospheres with $T_{\mathrm{eff}} \ll 10^7$ K (Z96).

Following these assumptions, a standard plane-parallel atmosphere model is used in conjunction with a radiative transfer calculation. The atmospheric modeling and radiative transfer follow the descriptions elsewhere (Mihalas 1970, 1978; Rybicki & Lightman 1979) and benefited greatly from the discussions of previous calculations (Z96; G02; H06).

Neglecting radiation pressure, the pressure at any depth is equal to the weight of the integrated column y of material above it:

$$P(y) = g_s\, y, \tag{1}$$

where g_s is the surface gravity, taken to be uniformly constant as per the plane-parallel atmosphere assumption. The surface gravity is assumed not to vary throughout the atmosphere since a negligible fraction of the neutron star mass is contained in the atmosphere, and its thickness is negligible compared to the neutron star radius.

The pressure is related to the temperature T and density ρ through the EOS, $P = P(\rho, T)$, which can be used to find the density at a given pressure (i.e., column depth) and temperature. To solve for the structure of the atmosphere, one additional expression is needed: the temperature as a function of column depth $T = T(y)$, which is referred to as the temperature profile. The temperature profile depends on the radiative transfer through the atmosphere. Thus, the temperature profile, atmospheric structure, and radiative transfer must be solved for jointly and iteratively.

Setting up the standard Feautrier problem (Feautrier 1964), the second-order form of the radiative transfer equation is

$$\mu^2 \frac{d^2 P_\nu(y,\mu)}{d\tau_\nu} = P_\nu(y,\mu) - S_\nu(y,\mu), \tag{2}$$

where τ_ν is the effective normal optical depth, μ is the (positive) cosine of the angle to the outward normal,

$$P_\nu(y,\mu) \equiv \frac{1}{2}\left[I_\nu(y,\mu) + I_\nu(y,-\mu)\right], \tag{3}$$

for outgoing ($I_\nu(y,\mu)$) and ingoing ($I_\nu(y,-\mu)$) specific intensity in directions given by μ, and $S_\nu(y, P_\nu)$ is the source function in those directions. The boundary conditions used are zero incident specific intensity at the surface (P_ν is half the outgoing specific intensity at the surface) and P_ν equal to the blackbody (BB) intensity B_ν at the deepest point considered. In terms of the true absorption opacity κ_ν and the reduced scattering opacity $\sigma_{\mathrm{T,r}}$, the source function can be expressed as

$$S_\nu(y,\mu) = \frac{\kappa_\nu}{\kappa_\nu + \sigma_{\mathrm{T,r}}} B_\nu(T(y)) + \frac{2}{\kappa_\nu + \sigma_{\mathrm{T,r}}} \int_0^1 P_\nu(y,\mu') \frac{d\sigma_{\mathrm{T,r}}}{d\mu'}(\mu,\mu') d\mu', \tag{4}$$

where $d\sigma_{\mathrm{T,r}}/d\mu'(\mu,\mu')$ is the differential amount of scattering from directions μ' into directions μ. Note that the present form of the source function differs from that used in previous works (Z96; G02; H06) since the scattering is not assumed to be isotropic. The source function depends on the temperature profile of the atmosphere, so the coupled radiative transfer and structure equations must be solved to calculate the emergent spectrum.

To solve Equation (2), the angle integral in the source function is discretized using a Gauss–Legendre quadrature sum with weights $a_{j'}$:

$$\int_0^1 P_\nu(y,\mu') \frac{d\sigma_{\mathrm{T,r}}}{d\mu'}(\mu,\mu') d\mu' \approx \sum_{j'=1}^{M} P_\nu(y,\mu_{j'}) \frac{d\sigma_{\mathrm{T,r}}}{d\mu'}(\mu,\mu_{j'}) a_{j'}\,. \tag{5}$$

Defining the dimensionless scattering albedo

$$\rho_\nu = \frac{\sigma_{\mathrm{T,r}}}{\kappa_\nu + \sigma_{\mathrm{T,r}}} = \frac{\sigma_{\mathrm{T,r}}}{k_\nu}, \tag{6}$$

where $k_\nu = \kappa_\nu + \sigma_{\mathrm{T,r}}$ is the total absorption opacity, allows Equation (2) to be rewritten as

$$\frac{\mu^2}{k_\nu}\frac{d}{dy}\left(\frac{1}{k_\nu}\frac{dP_\nu(y,\mu)}{dy}\right) = P_\nu(y,\mu) - (1-\rho_\nu)B_\nu(T(y)) - 2\rho_\nu \sum_{j'=1}^{M} P_\nu(y,\mu_{j'}) \frac{d\sigma_{\mathrm{T,r}}}{d\mu'}(\mu,\mu_{j'}) a_{j'}, \tag{7}$$

where use has also been made of the relation $d\tau_\nu = k_\nu dy$. This form of the radiative transfer equation can then be discretized to generate a system of equations that can be solved numerically.

For a neutron star with a homogeneous temperature distribution across the surface, the observed flux at a distance D is related to the emergent flux $F(R, \nu)$, where R is the neutron star radius, through the relation (McClintock et al. 2004)

$$F(D, \nu_{\rm obs}) = (1+z)^{-1}\frac{R^2}{D^2}F\left(R, (1+z)\nu_{\rm obs}\right), \tag{8}$$

where $\nu_{\rm obs}$ is the observed frequency. The redshift z is defined through

$$1+z = \frac{1}{\sqrt{1-\frac{2GM}{Rc^2}}}, \tag{9}$$

where G is Newton's constant and M is the mass of the neutron star.

3. METHOD OF SOLUTION

To solve the coupled radiative transfer and structure equations, an iterative scheme is adopted, similar to that used in a previous model (Z96). An initial guess at the temperature profile is required to calculate the initial radiative transfer, the solution to which is then used to calculate a correction to the temperature profile. The radiative transfer is then recalculated based on the corrected temperature profile, which allows a further correction to the temperature profile to be calculated. Iteration proceeds until the maximum fractional temperature correction computed for any depth falls below a chosen threshold (10^{-4} throughout the present work).

3.1. Initial Structure

The main inputs to the model are the target effective temperature, $T_{\rm eff}$, and the surface gravity g_s. The variable ν is discretized into K frequencies, logarithmically spaced between some minimum and maximum multiple of $k_{\rm B}T_{\rm eff}$ (0.05 and 120, respectively, throughout the present work). The variable y is discretized into N column depths, logarithmically spaced between a minimum and maximum column depth. If the largest column depth corresponds to an effective normal optical depth smaller than a specified minimum at the highest photon frequency considered, the maximum column depth is doubled until the optical depth is greater than the specified minimum ($\tau_{\nu\,\rm max} > 80$ is required throughout the present work, except where specified in Section 4.4). The variable μ is discretized into M angles, based on a Gauss–Legendre quadrature scheme.

The pressure at each column depth can be computed using Equation (1). In the Eddington approximation, the temperature profile satisfies the equation (Z96)

$$\frac{d}{dy}\frac{T}{T_{\rm eff}} = \frac{3}{16}k_{\rm R}\left(\frac{T}{T_{\rm eff}}\right)^3, \tag{10}$$

where $k_{\rm R}$ is the Rosseland mean total absorption opacity. This equation can be integrated from the surface to get the initial temperature profile, using the boundary condition $T(0) = 0.265T_{\rm eff}$. This boundary condition is found to be closer to the final converged value for hydrogen atmospheres than that used for this and other compositions in previous work (Z96). At each step in the integration, ρ is computed using OPAL EOS data (Rogers et al. 1996), and the Rosseland mean opacity $k_{\rm R}$ is computed from κ_ν and $\sigma_{\rm T,\,r}$.

3.2. Opacities and Ionization Fraction

For an ionization fraction $f_{\rm ion}$ at a given depth, the reduced Thomson scattering opacity is

$$\sigma_{\rm T,\,r} = \frac{f_{\rm ion}\,\sigma_{\rm T}}{f_{\rm ion}(m_p+m_e)+(1-f_{\rm ion})m_{\rm H}}, \tag{11}$$

where m_p, m_e, and $m_{\rm H}$ are the proton, electron, and hydrogen masses, respectively, and $\sigma_{\rm T}$ is the Thomson cross section.

If the hydrogen is completely ionized, the true absorption opacity is the free–free opacity (Rybicki & Lightman 1979),

$$\kappa_\nu = \frac{4e^6}{3m_e hc}\sqrt{\frac{2\pi}{3km_e}}T^{-1/2}\frac{\rho}{(m_p+m_e)^2}\nu^{-3} \times (1-e^{-h\nu/kT})\,\bar{g}_{\rm ff}\ {\rm cm^2\ g^{-1}}, \tag{12}$$

where $\bar{g}_{\rm ff}$ is the temperature-averaged free–free Gaunt factor, and all units are in cgs.

The temperature-averaged free–free Gaunt factor is commonly expressed in terms of the scaled frequency $u = h\nu/k_{\rm B}T$ and scaled temperature $\gamma^2 = Z^2{\rm Ry}/(k_{\rm B}T)$ and is available in tabulated form (Sutherland 1998).[2] At present, a two-dimensional linear interpolation of this data is performed to find the needed Gaunt factors, but this could be upgraded to a more accurate spline procedure. The tabulated data are available for u and γ^2 values in the range 10^{-4} to 10^4. In principle, it is possible to specify parameters that lead to u values outside this range in certain regions of the atmosphere. However, those regions are likely to be very deep in the atmosphere for the lowest frequencies, or near the surface at the highest frequencies. Low frequencies carry practically no flux deep in the atmosphere, while the upper layers of the atmosphere are transparent to high-frequency photons, so the Gaunt factor will not be significant in these regions. Thus, an adequate solution is to return the closest value on the boundary of the grid if a Gaunt factor is requested for parameters outside the grid.

For $T_{\rm eff} \geqslant 10^{5.3}$ K the atmosphere is highly ionized ($f_{\rm ion} > 0.95$; Z96), but neutral hydrogen could nevertheless dominate the opacity at some frequencies (especially near spectral lines, and for $T_{\rm eff}$ in the lower part of that range). To capture this effect, partial ionization can be enabled in McPHAC, which then obtains the ionization fraction and monochromatic opacities using tabulated data from the Opacity Project[3] (OP; Seaton et al. 1994). The opacities are valid for $\rho < 0.01\,{\rm g\,cm^{-2}}$, so a smooth transition from the OP opacities to the free–free opacity is used for $0.001\,{\rm g\,cm^{-2}} < \rho < 0.01\,{\rm g\,cm^{-2}}$. Further, a similar issue with the Gaunt factor tables arises, where u values outside the grid are requested for high frequencies at low temperatures. The opacities at these values are unlikely to be important in the radiative transfer but should fall off smoothly to avoid the

[2] The table was downloaded from http://www.mso.anu.edu.au/~ralph/data/freefree/gffgu.dat (last updated 2000 March 6). However, it appears that the column labels u and γ^2 are reversed, both in the online table and in the corresponding table in the paper, compared to what they should be. The data in the table only match the corresponding figure in the same paper if the labels are reversed. That figure appears to be in agreement with previous work (Karzas & Latter 1961), so it is most likely the tables that are mislabeled, rather than the figure. Further, two separate mentions of this issue appear elsewhere (Ogley et al. 2002, and http://hea-www.harvard.edu/PINTofALE/pro/external/sutherland_ch.pro), so the columns are assumed to have been mislabeled when reading the tabulated data.

[3] Data from OPCD_3.3 (Seaton 2005) are used, downloaded from http://cdsweb.u-strasbg.fr/topbase/op.html

possibility of interfering with convergence. Thus, the opacity at the largest tabulated value of u is extrapolated assuming the same frequency dependence as the free–free opacity. Finally, because the total opacity is tabulated, the derived true absorption opacity may not be accurate when this is smaller than the scattering opacity, so this is smoothly transitioned to the free–free value for absorption opacities smaller than the scattering opacity.

For full self-consistency the ionization fraction used in the opacities should be identical to that used in the OPAL EOS, but this is unlikely to be exactly satisfied for the OP opacities (or if enforcing complete ionization in the opacities). However, the effect of small amounts of neutral hydrogen on the EOS is expected to be much smaller than its effect on the opacities, so any such inconsistency is unlikely to be important for the effective temperatures considered in the present work.

3.3. Radiative Transfer

Having obtained an initial (or later, a corrected) atmospheric structure, the radiative transfer through the atmosphere must now be solved. The assumption of coherent scattering leads to an important computational simplification, in that the radiative transfer equation for a fixed atmospheric structure can be solved for one frequency at a time, since there is no direct frequency coupling in Equations (2) and (4). The only frequency coupling in the radiative transfer is through the effect of radiative transfer on the structure of the atmosphere, which will enter into the temperature correction procedure.

At a given frequency, there will be some maximum optical depth from beyond which there is not a significant fractional contribution to the flux at the surface. Thus, to calculate the flux at the surface, it is sufficient to consider column depths down to some maximum optical depth at that frequency. The maximum optical depth needed for a specific fractional flux contribution at the surface will vary with the frequency considered and the structure of the atmosphere, but in the present work a single prespecified maximum effective normal optical depth (= 80) is used for each frequency.

For each frequency, the column depth is discretized into $N_{\rm rad}$ depth points between the minimum column depth and the first column depth corresponding to an effective normal optical depth greater than the specified maximum. The temperature profile is then interpolated to these depth points, and the density and opacities computed there, allowing the radiative transfer problem to be solved on this frequency-specific grid of column depths and angles.

For the fixed atmospheric structure, the radiative transfer equation (Equation (2), with the specific source function given in Equation (4)) is solved using a Hermitian method described completely in previous work (Auer 1976). The general form of the transfer equation solved in that work is

$$\mu_j^2 \frac{d^2 P_j}{d\tau_j^2} = \Sigma_{j'=1}^{N} R_{j,j'} P_{j'} - T_j, \tag{13}$$

where j and j' are discrete frequency–angle points. To apply that method of solution to the present problem, the specific forms of $R_{j,j'}$ and T_j must be given, as well as the explicit boundary conditions. In the present application (see Equation (7)),

$$R_{j,j'} = 1 - \frac{2}{\kappa_\nu + \sigma_{\rm T,r}} \frac{d\sigma_{\rm T,r}}{d\Omega}(\mu_j, \mu_{j'})\, a_{j'} \tag{14}$$

and

$$T_j = \frac{\kappa_\nu}{\kappa_\nu + \sigma_{\rm T,r}} B_\nu\left(T(y)\right), \tag{15}$$

where $a_{j'}$ is the Gauss–Legendre quadrature weight for the angle $\mu_{j'}$. Using the phase function for dipole scattering (Hummer 1962), the differential amount of scattering from directions μ' into directions μ can be expressed as

$$\frac{d\sigma_{\rm T,r}}{d\mu'}(\mu_j, \mu_{j'}) = \sigma_{\rm T,r} \frac{3}{16}\left(3 + 3\mu_j^2\mu_{j'}^2 - \mu_j^2 - \mu_{j'}^2\right). \tag{16}$$

The boundary condition at the surface is simply $I_- = 0$, i.e., no incident specific intensity. For the bottom boundary condition, the second-order accurate expression $P_\nu = B_\nu$ is used, which is an approximation to the third-order accurate diffusion approximation boundary condition suggested in previous work (Auer 1976). The reason for choosing this specific bottom boundary condition is that it ensures that the specific intensity is isotropic at the bottom depth considered at each frequency, which will be useful when computing the Eddington factors in the temperature correction scheme.

Having specified $R_{j,j'}$, T_j, and the boundary conditions, the radiative transfer problem can be solved using the method described in previous work (Auer 1976), giving $P_\nu(y, \mu)$ on the grid of depth and angle points used for each specific frequency. The emergent specific intensity at the surface is then $I_\nu(0, \mu) = 2P_\nu(0, \mu)$.

Under the isotropic scattering assumption the phase function is $g = 1/2$ (Hummer 1962), and the expression for the differential amount of scattering becomes

$$\frac{d\sigma_{\rm T,r}}{d\mu'}(\mu_j, \mu_{j'}) = \sigma_{\rm T,r} \frac{1}{2}. \tag{17}$$

The anisotropic expression is used throughout the present work, except in Section 4.2, where the spectra computed with the anisotropic scattering expression are compared to those computed with the isotropic scattering expression, and in Section 4.4, where the isotropic expression is used in order to match the assumptions of the previous work (Z96).

3.4. Temperature Correction

The temperature correction scheme used is the same as that employed in a previous work (Z96), though a different discretization method is used. Both the temperature correction procedure and discretization method are described in full in Appendix A. The temperature correction procedure requires the solution of the equation

$$\frac{1}{k_\nu}\frac{d}{dy}\left(\frac{1}{k_\nu}\frac{d}{dy}(f_\nu J_\nu)\right) = \left(J_\nu - B_\nu(T_0) - \frac{dB_\nu}{dT}\frac{\int_0^\infty (J_\nu - B_\nu(T_0))\kappa_\nu d\nu}{\int_0^\infty \frac{dB_\nu}{dT}\kappa_\nu d\nu}\right)\frac{1}{k_\nu}, \tag{18}$$

where

$$f_\nu = J_\nu^{-1}\int_0^1 \mu^2 P_\nu(y, \mu) d\mu \tag{19}$$

is one of the Eddington factors and

$$J_\nu = \int_0^1 P_\nu(y, \mu) d\mu \tag{20}$$

is the mean specific intensity. Note, however, that this is the mean specific intensity subject to the constraint of constant

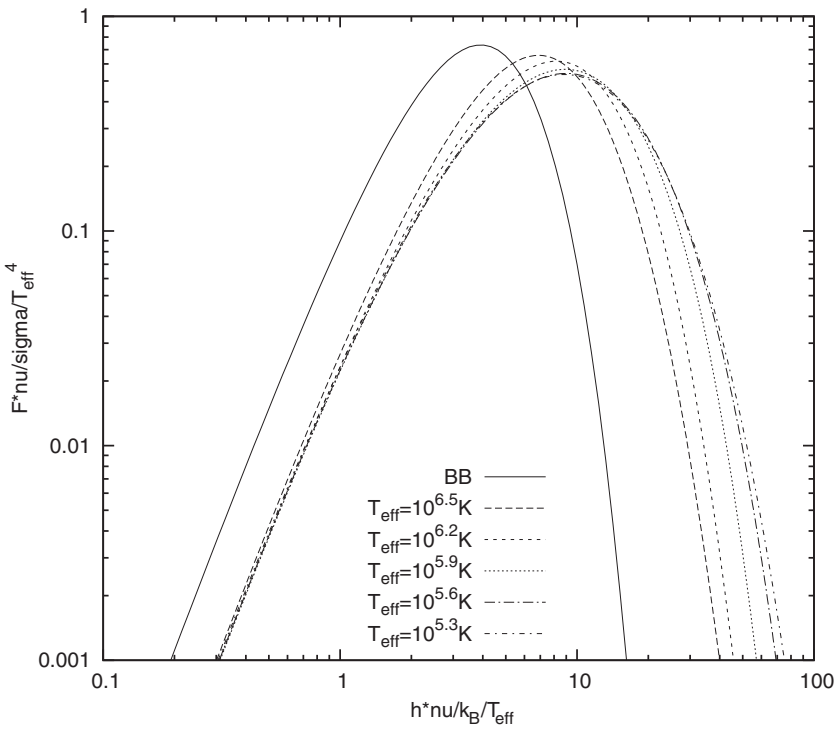

Figure 1. Normalized spectra computed (assuming complete ionization) for various effective temperatures, along with a blackbody (BB) spectrum for comparison. The hottest of the computed spectra is the closest to the blackbody spectrum, and the spectra get harder with decreasing effective temperature.

energy flux through the atmosphere, which will only equal the mean specific intensity from the radiative transfer calculation in the limit of zero temperature correction. The boundary condition at the surface is

$$\frac{1}{k_\nu}\frac{d}{dy}(f_\nu J_\nu) = h_\nu J_\nu, \tag{21}$$

where

$$h_\nu = J_\nu^{-1}\int_0^1 \mu P_\nu(y,\mu)d\mu \tag{22}$$

is another Eddington factor, and the boundary condition at the largest column depth is $J_\nu = B_\nu$.

Note that allowing for anisotropic scattering in the present work does not change the form of Equation (18), since the angular integral over the source function must come out to the same whether or not it has an angular dependence (all radiation "absorbed" due to scattering opacity must be "re-emitted").

The temperature correction $\Delta T(y)$ is equal to the ratio of the two integrals in the last term of Equation (18) (see Equation (A5)). To find J_ν, and thus ΔT, Equation (18) is solved using the f_ν computed in the radiative transfer problem. For the depth points beyond the maximum column depth considered at each frequency in the radiative transfer problem, the isotropic limit $f_\nu = 1/3$ is used. A continuous f_ν is ensured by the use of the boundary condition $P_\nu = B_\nu$ in the radiative transfer problem.

Using a fixed $f_\nu(y)$, Equation (18) can be solved using a Rybicki-type solution (Rybicki 1971), the details of which are given in Appendix A. The resulting temperature correction is applied to the temperature profile, and the radiative transfer repeated to give an updated $f_\nu(y)$, which in turn is used to correct the temperature profile. Iteration is continued until the maximum computed fractional temperature change is less than a specified threshold (10^{-4} throughout the present work).

4. RESULTS

This section presents results obtained with McPHAC v1.1. The aim is to present the capabilities and limitations of the present version of the code, to inspire future improvements and applications. Thus, previous works (Z96; G02; H06) are referred to for a more complete discussion of the hydrogen atmosphere structure and emergent spectrum.

The code has been compiled and run on an AMD platform running the Linux operating system. For details on the code implementation and numerical packages used, see Appendix B.

Unless otherwise stated, the following "default" level of discretization is used: the number of frequency points used is $K = 100$, the number of angle points used is $M = 40$, the number of depth points used in the radiative transfer calculation is $N_{\rm rad} = 3200$, and the number of depth points used in the temperature correction procedure is $N_{\rm temp} = 3200$. Further, the minimum column depth considered at $T_{\rm eff} = 10^{6.5}$ K is $y_{\rm min} = 10^{-5}\,{\rm g\,cm^{-2}}$, while it is $y_{\rm min} = 10^{-9}\,{\rm g\,cm^{-2}}$ at all lower effective temperatures. For all effective temperatures the maximum column depth is set to $y_{\rm max} = 10^2\,{\rm g\,cm^{-2}}$, though this may be increased by the code as described in Section 3.1.

For Sections 4.1–4.3 and 4.5, complete ionization is assumed when computing the opacities, while the effect of partial ionization is included in Section 4.4 for a better comparison with the previous work. As previously noted, there is a slight inconsistency in assuming complete ionization for the opacities but not for the EOS, but this is of minor importance in characterizing the computational capabilities of the code, which is the aim of this section.

4.1. Emergent Spectra

Normalized and rescaled spectra computed (assuming complete ionization) for various effective temperatures are shown in Figure 1, along with a BB spectrum for comparison. The

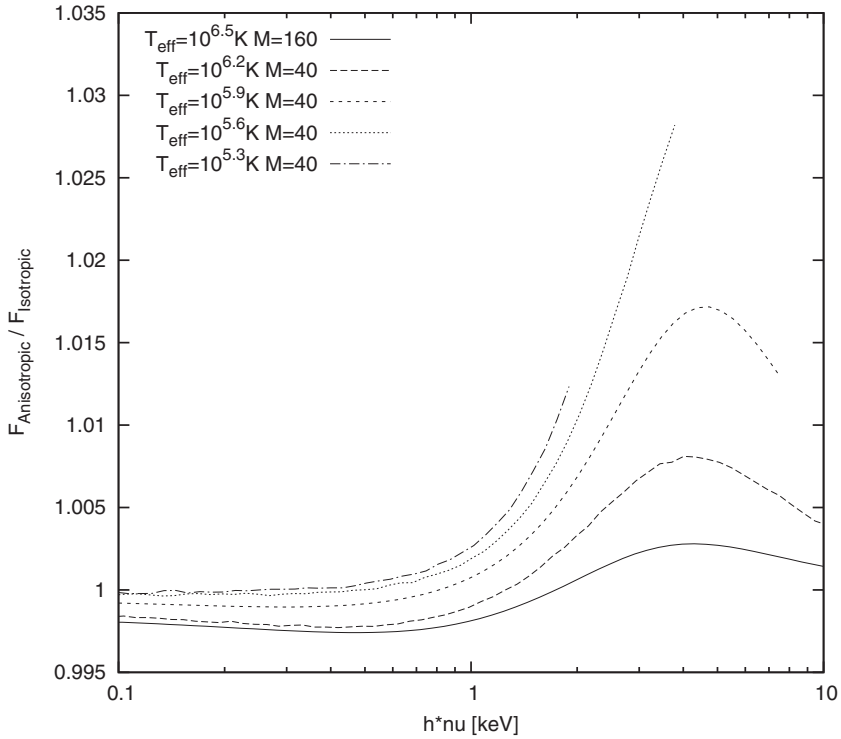

Figure 2. Ratio of the spectrum calculated considering anisotropic Thomson scattering to that computed assuming isotropic scattering, for various effective temperatures. The curves terminate at $h\nu = 120 k_B T_{eff}$, which corresponds to the highest frequency considered. Considering anisotropic scattering appears to increase the flux above 1 keV, with the largest increase happening around 5 keV (apparently insensitive to effective temperature). The effective temperatures of each pair of spectra differ by less than 2×10^{-5}. Thus, increasing the flux above 1 keV reduces the flux below 1 keV to keep the effective temperature the same.

default discretization level described previously is used, with the exception of the $T_{eff} = 10^{6.5}$ K spectrum, for which $M = 160$ angle points were used. The hottest of the computed spectra lies the closest to the BB spectrum, and the spectra get harder with decreasing effective temperature. Thomson scattering is expected to be more important in the hotter atmospheres, and the relative magnitudes of the free–free and reduced Thomson opacities as a function of frequency appear to be important in determining the shape of the spectrum. The spectra are significantly harder than a BB spectrum because of the frequency dependence of the free–free opacity (the free–free cross section decreases with increasing frequency), which means that the flux at a higher frequency originates deeper in the atmosphere, where the temperature is greater. For higher effective temperatures, Thomson scattering is more important since the spectrum is shifted to higher energies, thus reducing the free–free opacity. Since the reduced Thomson opacity is frequency independent, it softens the spectrum by limiting the depth in the atmosphere from which the high-energy side of the spectrum originates. The behavior described here is similar to that seen in previous work (H06), where the high-energy side of the rescaled spectrum steepened with increasing effective temperature.

4.2. Anisotropic Thomson Scattering

The present work allows for Thomson scattering to be treated anisotropically, in contrast to previous work (Z96; G02; H06). The effect of relaxing the isotropy assumption is investigated here, by comparing the spectra computed with the McPHAC code using the anisotropic scattering expression to those computed using the isotropic scattering expression (see Section 3.3).

Figure 2 shows the ratio of the spectrum calculated considering anisotropic Thomson scattering to that computed assuming isotropic scattering, for various effective temperatures (assuming complete ionization). Considering anisotropic scattering appears to increase the flux above 1 keV, with the largest increase happening around 5 keV (apparently insensitive to effective temperature). The effective temperatures of each pair of spectra differ by less than 2×10^{-5}. Thus, increasing the flux above 1 keV reduces the flux below 1 keV to keep the effective temperature the same. Corrections to anisotropic Thomson scattering due to collective effects (Boercker 1987) may change the importance of considering anisotropic scattering, but it appears that isotropic scattering is an adequate assumption as long as one is not seeking accuracy better than 1% in flux as a function of photon energy.

4.3. Convergence of Computational Method

There are multiple sources of uncertainty in the spectra computed under the assumptions described in Section 2: uncertainty in the input parameters (e.g., opacities or EOS), truncation (discretization) uncertainty in the numerical method (for both the temperature correction and radiative transfer), and the convergence uncertainty between iterations. Investigation of the effect of uncertainty in the input parameters is left for future work, as the aim of the present work is limited to presenting the computational capabilities of the McPHAC code, while the latter two forms of uncertainty are investigated here.

The truncation error is the error introduced by truncating the power series describing a function to a finite order. The local truncation error from one point to another can usually be constrained but the additive effects of truncation errors are more difficult to quantify. One way to estimate the size

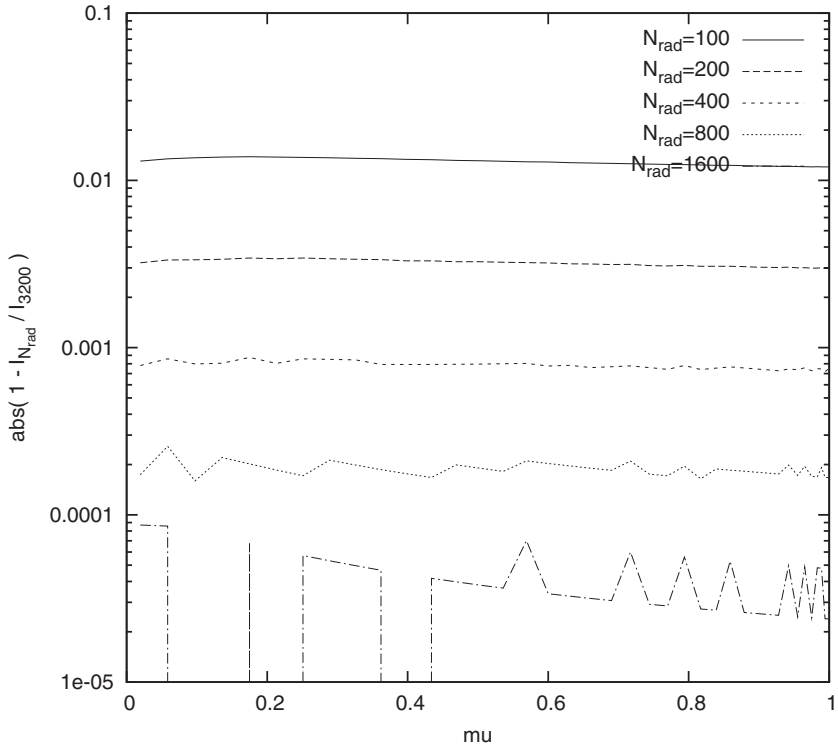

Figure 3. Convergence of the specific intensity resulting from repeating the radiative transfer calculation for an increasing number of depth points. The effective temperature of the spectrum is $T_{\rm eff} = 10^{6.5}$ K, and the specific intensity shown is for a frequency close to the spectrum peak (frequency number $k = 3K/5$). Note that the convergence scales approximately as $1/N_{\rm rad}^2$, and that doubling $N_{\rm rad}$ thus decreases by a factor of four the fractional difference between the specific intensity computed for $N_{\rm rad}$ depth points and that computed using 3200 depth points.

of the truncation error is to perform the calculation for an increasingly fine discretization and to analyze the convergence of the resulting value of interest. In general, the truncation error in each equation due to the discretization of each variable should be investigated, so a multidimensional grid of discretization levels should be used for each of the input parameters for which spectra are computed. However, at present a more limited approach is taken, intended mainly to give a sense of the capabilities and limitations of the code. A more thorough treatment should be performed in specific applications where it is important to constrain the uncertainty in the computed spectra.

The discretization of column/optical depth in the radiative transfer calculation is perhaps the most obvious example of how a truncation error can be introduced in the computed spectrum. To constrain this, the radiative transfer calculation was repeated for a fixed atmospheric structure, varying $N_{\rm rad}$ while keeping all other discretizations at their default values. Figure 3 shows the convergence of the computed specific intensity at the surface with increasing $N_{\rm rad}$, for a frequency close to the peak of the $T_{\rm eff} = 10^{6.5}$ K spectrum. The specific intensities shown are those computed after six iterations, where the maximum fractional temperature change in the atmosphere in the last iteration is less than 10^{-4}. As the number of depth points in the radiative transfer calculation is increased from 100 to 1600, the fractional difference in the computed spectra falls below 10^{-4}, indicating that the truncation (discretization) error in the radiative transfer calculation due to discretizing the depth variable is smaller than 10^{-4} when using 3200 depth points. Hardly any dependence on μ is seen in the convergence, indicating that convergence can be demonstrated separately for the discretizations in μ and in depth, rather than resorting to a multidimensional grid of discretization levels. Further, this indicates that a single component may limit the convergence at every angle. Identifying this component and finding some way to reduce it could improve convergence. However, even limited by this hypothetical component, second-order convergence is observed. The worst convergence is seen for the highest frequency considered, but even at that frequency second-order convergence to better than 10^{-4} is achieved using 3200 depth points.

Another truncation error that may be important in the radiative transfer calculation is that introduced through the discretization of the angle variable μ. To constrain just the truncation error, the radiative transfer calculation could be repeated for an increasing number M of angle points at fixed atmospheric structure. However, instead the entire iterative solution has been repeated, which captures all the effects of the discretization of μ on the emergent spectrum. This includes the effect of uncertainty in the computed Eddington factor on the computed temperature correction, though being simply a Gauss–Legendre approximation to an integral, the Eddington factor should converge quickly with increasing M. The convergence of the computed spectrum with number of angle points is shown in Figure 4, for $T_{\rm eff} = 10^{6.5}$ K. The spectra shown are those computed after six iterations, where the maximum fractional temperature change in the atmosphere in the last iteration is less than 10^{-4}. From the rate of convergence seen, it is expected that the uncertainty in the spectrum due to discretizing the μ variable is less than 10^{-4} when using 160 angle points (for this effective temperature).

As is seen in Figure 4, there is a sign change of $1 - F_{\rm M}/F_{160}$ near $h\nu/k_{\rm B}/T_{\rm eff} \sim 3.5$ for all values of M, indicating that the discretization of μ leads to a consistent underprediction of the flux below that frequency and a consistent overprediction of the flux above that frequency. The location of the sign change

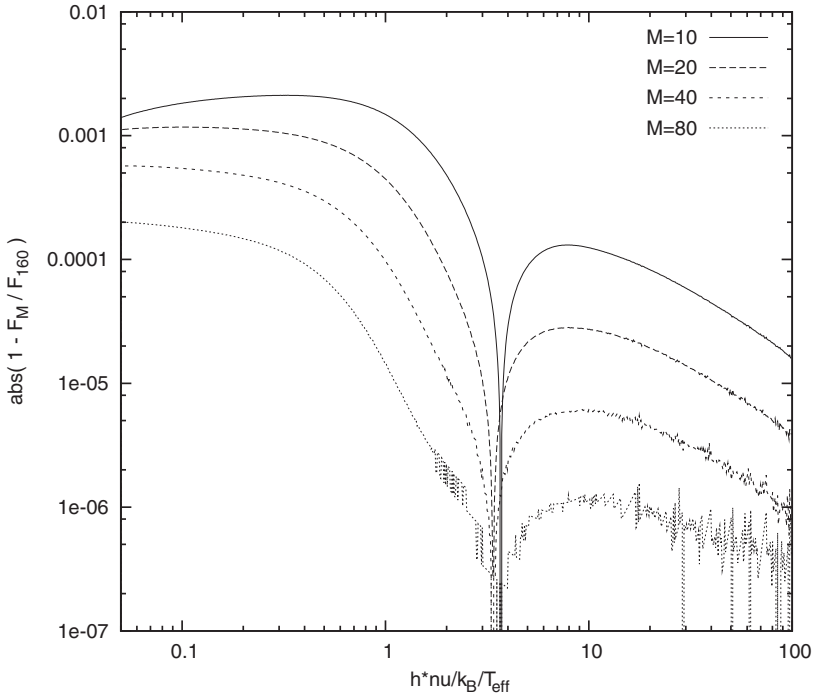

Figure 4. Convergence of the emergent spectrum resulting from re-running the full McPHAC code for an increasing number of angle points. The effective temperature of the spectrum is $T_{\rm eff} = 10^{6.5}$ K. There is a sign change of $1 - F_{\rm M}/F_{160}$ near $h\nu/k_{\rm B}/T_{\rm eff} \sim 3.5$ for all values of M, indicating that the discretization of μ leads to a consistent underprediction of the flux below that frequency and a consistent overprediction of the flux above that frequency.

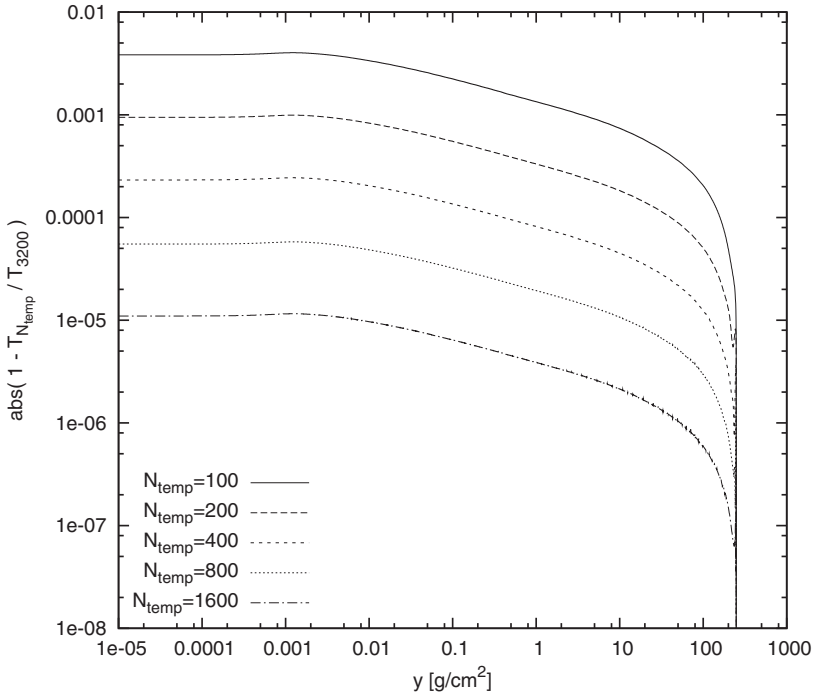

Figure 5. Convergence of the temperature profile resulting from repeating the temperature correction procedure with an increasing number of depth points. The flux through the atmosphere corresponds to $T_{\rm eff} = 10^{6.5}$ K. The first iteration is shown, since this is where the largest temperature correction is computed, but the corresponding figures for later iterations look almost identical. Note that the convergence scales approximately as $1/N_{\rm temp}^2$, and that doubling $N_{\rm temp}$ thus decreases by a factor of four the difference between the temperature profile computed with $N_{\rm temp}$ depth points and that computed with 3200 depth points. The sharp drop-off around $y = 300$ g cm^{-2} is due to the bottom boundary condition, which results in zero computed temperature correction there.

is close to the peak of the BB spectrum, suggesting that the different character of the radiation in the Rayleigh–Jean and Wien tails is a possible cause. The low-frequency parts of the emergent spectrum originate near the surface of the atmosphere and are reasonably isotropic, while the high-frequency parts of the spectrum originate deeper in the atmosphere and are more anisotropic. Nonetheless, using a sufficiently large M ensures an acceptably small fractional error in the computed emergent spectrum.

In calculating the temperature correction at each iteration, the truncation error due to discretizing the depth variable may be important. To investigate this, the temperature correction at

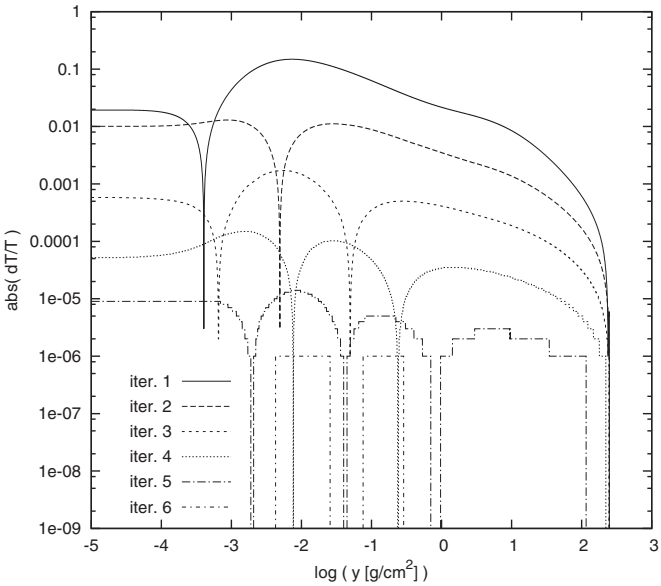

Figure 6. Computed fractional temperature change (ΔT, see Equation (A5)) with depth, for each iteration. The flux through the atmosphere corresponds to $T_{\rm eff} = 10^{6.5}$ K. The sharp decrements (e.g., near $\log(y) = -3.5$ for iteration 1) are due to changes in sign of ΔT.

each iteration is repeated for a varying number of depth points. The convergence of the temperature profile (after temperature correction) computed with $N_{\rm temp}$ depth points is shown in Figure 5, for a $T_{\rm eff} = 10^{6.5}$ K atmosphere. The first iteration is shown, since this is where the largest temperature correction is computed, but the corresponding figures for later iterations look almost identical. For anything more than 800 depth points, the truncation uncertainty in the temperature correction procedure introduced by discretizing y is less than 10^{-4} (at this effective temperature).

Separate from the truncation uncertainty, there is convergence uncertainty. The convergence uncertainty is the uncertainty introduced by solving coupled equations iteratively, since the solution at a given iteration is different from that which would be obtained in the limit of an infinite number of iterations. The convergence uncertainty can be constrained by analyzing the convergence of the resulting values for successive iterations.

The temperature profile is corrected at each iteration, and these corrections affect the structure of the atmosphere and the emergent spectrum. Thus, it is important that enough iterations are used to bring the largest fractional temperature change below some threshold. The convergence of the computed fractional temperature change with iteration is shown in Figure 6, for a $T_{\rm eff} = 10^{6.5}$ K atmosphere. At five iterations, the maximum computed fractional temperature change is below 10^{-4}. However, it is not obvious how the temperature corrections affect the emergent spectrum, so it is also verified that the emergent spectrum is sufficiently converged. At five iterations, the fractional convergence uncertainty in the specific intensity at the surface is less than 10^{-4} (for this effective temperature).

The most intuitive way to calculate the outward flux at the surface (the emergent spectrum) is the integral $F_\nu = \int_0^1 \mu\, I_\nu(\mu)\, d\mu$. However, it can also be computed from the Eddington factor f_ν and mean specific intensity J_ν through the expression (Z96)

$$F_\nu = \frac{4\pi}{\alpha_\nu + \sigma_\nu} \frac{d}{dy} (f_\nu\, J_\nu)\,. \tag{23}$$

Interestingly, J_ν is calculated in the temperature correction procedure, and in the limit of zero computed temperature correction, that J_ν is the same as that computed in the radiative transfer calculation. Thus, if the Eddington factors can be computed to sufficient accuracy in a computationally efficient way, it is possible that it is computationally advantageous to use the spectrum calculated from the temperature correction procedure, rather than to require sufficient accuracy of the calculated spectrum in the radiative transfer calculation. Figure 7 compares the normalized spectrum computed from the temperature correction procedure to that computed using the radiative transfer calculation. The spectra shown are those computed after six iterations, where the maximum fractional temperature change in the atmosphere in the last iteration is less than 10^{-4}. The number of angle points M used for each curve is indicated in the legend. The two curves (40 and 160 angle points) for $T_{\rm eff} = 10^{6.5}$ K are almost identical and show that the spectra computed using the two different methods differ by less than 10^{-4} for $h\,\nu/(k_{\rm B}\, T_{\rm eff}) < 50$. This indicates that the difference between the spectra computed with the radiative transfer calculation and with the temperature correction procedure is not sensitive to the number of angle points used. Further, the spectrum computed with the temperature correction procedure can be made accurate enough to be used in place of that computed in the radiative transfer calculation. The difference between the two spectra at the highest frequencies may be due to there being a significant contribution to the flux at the surface from beyond an effective normal optical depth of 80 at these frequencies, since this flux would be ignored in the radiative transfer calculation. The curve for $T_{\rm eff} = 10^{5.6}$ K shows a larger difference between the spectra computed with the two different methods, which may be due to the larger logarithmic range of column depths used at this effective temperature leading to a larger truncation error on the spectrum computed in the temperature correction procedure (since the number of depth points is fixed).

4.4. Comparison with Previous Work

For the purposes of comparison with previous work (Z96), spectra were produced with McPHAC including the effects of partial ionization on the opacities (see Section 3.2). The discretization level used was $K = 200$, $M = 10$, $N_{\rm rad} = 200$, and $N_{\rm temp} = 200$, intended to give computational uncertainty well below a percent. The range of column depths was in each case chosen to match that in previous work, though the maximum column depth was increased by the code as described in Section 3.1, except where specified in Figure 10.

Figure 8 shows a comparison of the normalized spectra computed using the McPHAC code with those of previous work (Z96), for various effective temperatures. There appears to be generally good agreement between the spectra, though the spectra computed in previous work predict significantly less flux at high energies than those of the present work for the three lowest effective temperatures.

Figure 9 shows the ratio of the pairs of spectra shown in Figure 8. The spectra are interpolated to allow for comparison. All the curves show a sharp drop-off at the high-frequency end, as well as an enhancement of several percent just before this drop-off. Near their peaks, the spectra computed using the McPHAC code agree with the spectra of previous work (Z96) to $\lesssim$1%. Little flux is expected at the energies above the drop-off, so this difference is unlikely to have practical consequences for most applications.

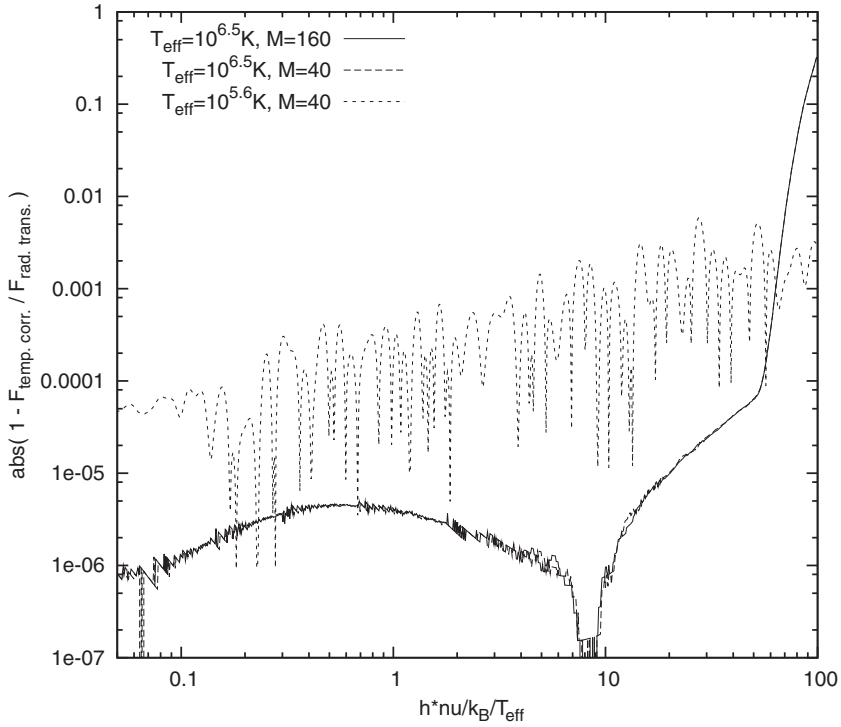

Figure 7. Comparison of the emergent spectrum resulting from the radiative transfer calculation to that resulting from the temperature correction procedure. The number of angle points used for each curve is indicated in the legend. The two curves (40 and 160 angle points) for $T_{\rm eff} = 10^{6.5}$ K are almost identical.

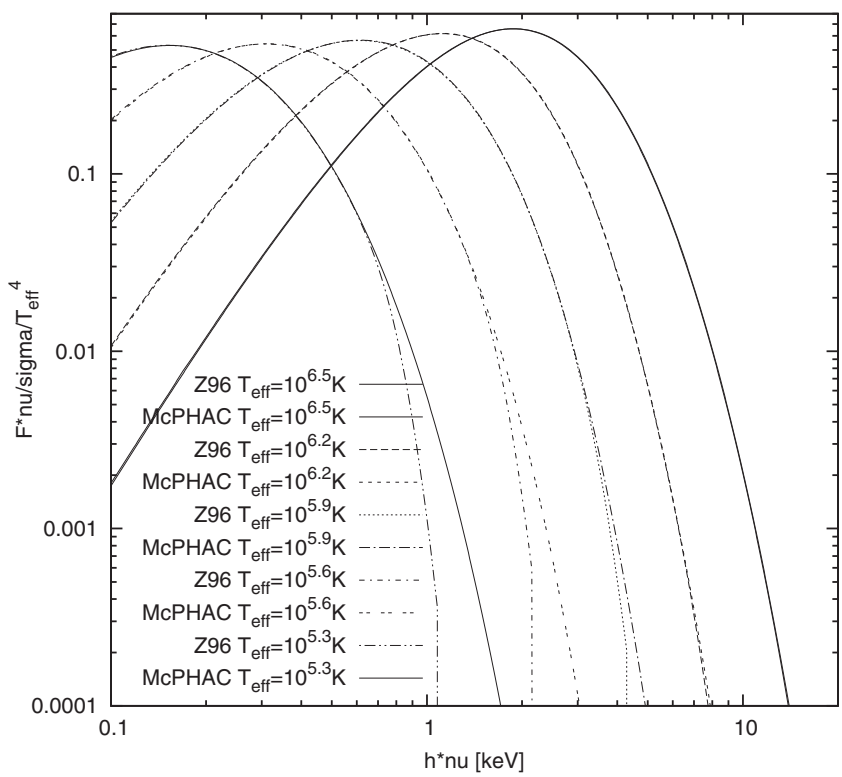

Figure 8. Comparison of the normalized spectra computed (considering partial ionization) using McPHAC with those of previous work (Z96), for various effective temperatures. The vertical high-frequency cutoffs seen for the three lowest effective temperature spectra of the previous work are present because the tabulated spectra are not specified at frequencies above the cutoff value. There generally is good agreement between the McPHAC spectra and those of the previous work, though the McPHAC spectra predict significantly more flux at high energies than those of the previous work for the three lowest effective temperatures.

One possible cause for the lower flux seen at the highest energies in the spectra of previous work (Z96) is that the previous work considered a smaller maximum column depth than the present work. To investigate this hypothesis, the McPHAC code was run for the column depth ranges used in the previous work for three of the effective temperatures (read from a figure in the previous work). Figure 10 shows the normalized spectrum of the previous work, the McPHAC

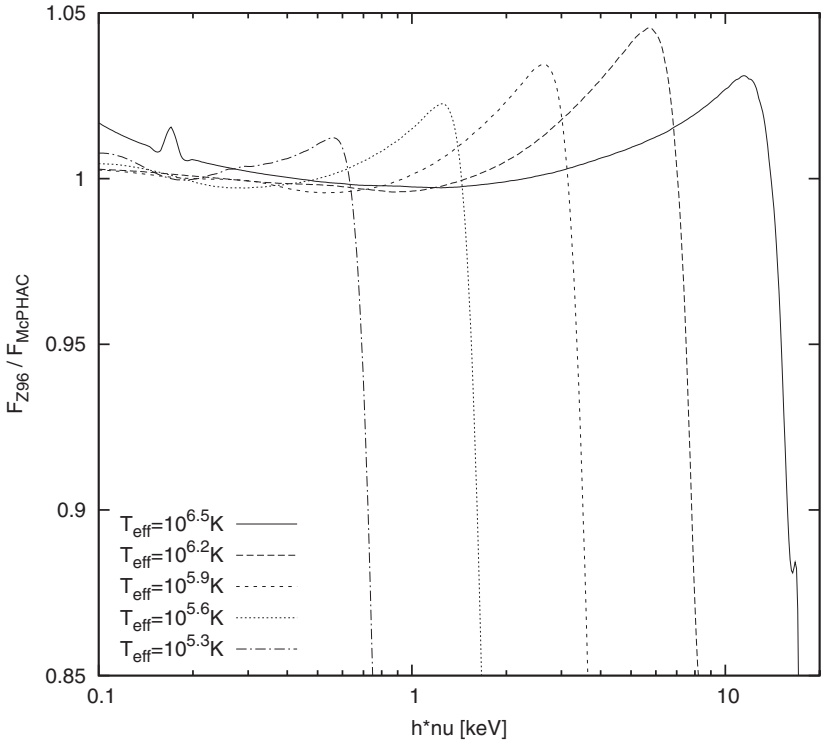

Figure 9. Ratio of the pairs of spectra shown in Figure 8. All the curves show a sharp drop-off at the high-frequency end, as well as an enhancement of several percent just before this drop-off. Near their peaks, the spectra computed using the McPHAC code agree with the spectra of previous work (Z96) to $\lesssim$1%.

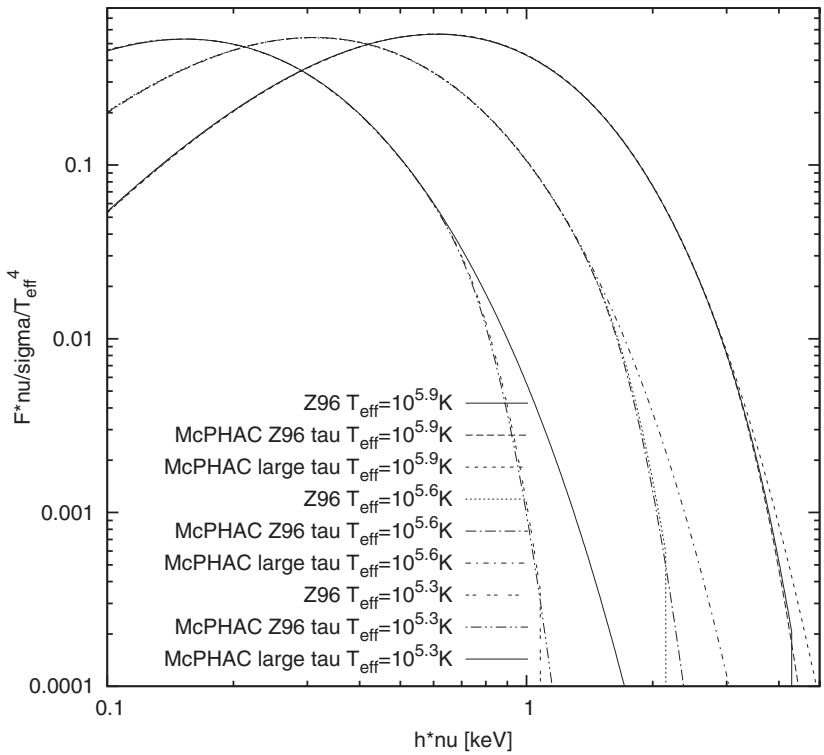

Figure 10. Comparison of the normalized spectra computed using McPHAC to those of previous work (Z96), for various effective temperatures. The column depth range used for the McPHAC spectra labeled "Z96 tau" at each effective temperature is taken to be the same as that used in the previous work. The column depth range for the McPHAC spectra labeled "large tau" ensures that the largest effective normal optical depth is at least 80 at all frequencies. The "Z96 tau" McPHAC spectra agree well with those of the previous work, indicating that the difference between these and the "large tau" McPHAC spectra is due to the previous work not considering large enough optical depths at the highest frequencies.

computed spectrum using the code-adjusted column depth range ("large tau"), and the McPHAC computed spectrum using the column depth range of the previous work ("Z96 tau"), for the three effective temperatures where a clear difference in the spectra was seen in Figure 8. The "Z96 tau" McPHAC spectra agree well with those of the previous work,

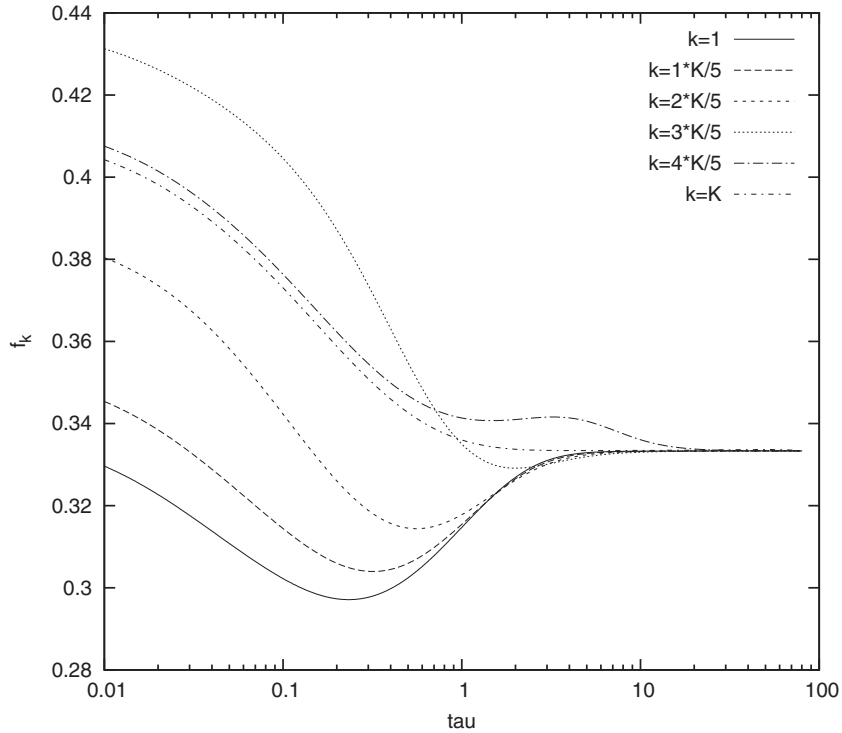

Figure 11. Eddington factor f, computed (assuming complete ionization) for a $T_{\rm eff} = 10^{6.5}$ K atmosphere, where k is the frequency number. The largest deviation from $f = 1/3$ occurs near the surface, for the peak frequency of the spectrum. For frequencies above the peak $f \gtrsim 1/3$, while for frequencies below the peak there are optical depths at which $f < 1/3$.

indicating that the difference between these and the "large tau" McPHAC spectra is due to the previous work not considering a large enough effective normal optical depth at the highest frequencies.

A fractional comparison between the "Z96 tau" spectra and those of previous work would not be illuminating since any differences in the high-frequency part of the spectrum could be attributed to not using the exact same column depth ranges, and any differences in the low-energy part of the spectrum are already shown in Figure 9. However, the discovery that the previous work did not consider a large enough maximum column depth does offer a possible explanation for the excess flux at energies just below the sharp drop-off seen in Figure 9. The temperature correction scheme used in the previous work attempts to enforce constant total radiative flux through the atmosphere, so it may be that the missing flux near the deepest column depths led to modifications to the temperature profile there that gave rise to the additional flux seen at frequencies just below the drop-off. This has not been investigated, however, and remains pure speculation.

4.5. Eddington Factors

The Eddington factor $f_\nu(\tau_\nu)$, computed (assuming complete ionization) for a $T_{\rm eff} = 10^{6.5}$ K atmosphere using $M = 160$ angle points, is shown in Figure 11. For isotropic specific intensity, the Eddington factor takes the value $f = 1/3$. If the specific intensity is larger for small μ than for large μ, then $f < 1/3$. Conversely, if the specific intensity is larger for large μ than for small μ, then $f > 1/3$. As expected, f goes to one-third at the largest optical depth, where the radiation is expected to be practically isotropic. The largest deviation from $f = 1/3$ occurs near the surface, for the peak frequency of the spectrum. For frequencies above the peak $f \gtrsim 1/3$, while for frequencies below the peak there are optical depths at which $f < 1/3$. A possible explanation for this behavior is as follows. The specific intensity for small μ is set mainly by the BB function for the temperature at that depth. Since the temperature increases monotonically with increasing τ_ν, so does the specific intensity for small μ. The ingoing specific intensity for large μ depends on the amount of emitting material above the depth considered and on the temperature and opacity of that material. Thus, the ingoing specific intensity for large μ should increase more slowly with optical depth than the specific intensity for small μ. The outgoing specific intensity for large μ should not change by much in the top fraction of an optical depth of the atmosphere, so the Eddington factor should thus initially decrease with increasing optical depth from the surface. The magnitude of the outgoing specific intensity at large μ compared to the specific intensity at small μ determines whether there is a region where $f < 1/3$ for a given frequency. For frequencies above the spectrum peak, the outgoing specific intensity for large μ appears to be large enough to keep $f \gtrsim 1/3$, while this is not the case for frequencies below the peak. The graph for the Eddington factor h looks similar, except h is $1/2$ for isotropic specific intensity, not $1/3$. Only the value of h at the surface is used, however, so the depth dependence is not really of interest.

5. FUTURE IMPROVEMENTS

In this section, brief discussion of prospective modifications to the code is given. These modifications could be made in the future to improve McPHAC functionality, its applicability to a particular analysis, or its efficiency.

Collective effects. Corrections to the Thomson scattering opacity due to collective effects (Boercker 1987) should be included. This has been partially implemented based on code from the OPCD_3.3 but should be completed and tested.

Electron heat transport. Zero electron heat transport is an assumption in the present version of the code. The code could be generalized to allow for electron heat transport, which may affect the atmospheric structure enough to significantly affect the emergent spectra. This may also produce a surface-gravity-dependent effect.

Updating the EOS and examining the effect of uncertainties in the EOS. The OPAL EOS functions are used to get the EOS of the atmosphere. The code should be updated to use the most recent version of these functions, and the effect of any uncertainties in the EOS should be investigated, as should the range of validity in temperature and pressure of the functions.

Improved opacities for partially ionized hydrogen. The OPCD_3.3 monochromatic opacities and ionization fractions are tabulated for a limited range of densities at each temperature and could be upgraded to include more complete tables.

Include Comptonization. Scattering is at present assumed to be coherent, but the effects of Comptonization should ultimately be included in McPHAC. Comptonization is likely to be important for H atmospheres with very high effective temperature.

Determine validity of radiative transfer. The equations of radiative transfer may not be applicable in deep layers of the atmosphere because the plasma frequency may approach the radiation frequency. The validity of the radiative transfer equations in these regions should therefore be assessed.

Set initial atmosphere structure based on previously computed atmosphere for similar parameters. For computing emergent spectra on a grid of effective temperatures and surface gravities, it is likely that great computational gains could be made by setting the initial atmospheric structure based on an atmosphere with similar parameters. Doing so could reduce the number of iterations needed for satisfactory convergence.

Use a Rybicki-type solution for the case of isotropic Thomson scattering. A Rybicki-type modification (Rybicki 1971) to the Feautrier solution cannot be used in McPHAC for the case of anisotropic Thomson scattering but is possible under the isotropic scattering approximation. A Rybicki-type solution would likely be less computationally demanding and may therefore be preferable in cases where it is not necessary to treat Thomson scattering anisotropically. The code to use a Rybicki-type solution in the case of isotropic scattering is largely implemented but remains untested. It should also be noted that the Rybicki solution allows coupling between frequencies in the radiative transfer calculation and thus allows incoherent scattering.

Solve full Feautrier problem using sparse matrix methods. Modern iterative techniques allow for the solution of very large sparse matrix equations, so if it were desired to modify McPHAC in a way that would require the solution of the full Feautrier problem (arbitrary angle and frequency coupling allowed), such methods could be used to allow useful frequency, depth, and angle resolution.

6. DISCUSSION AND CONCLUSIONS

The McPHAC v1.1 code is presented, marking the beginning of an effort to produce model hydrogen atmosphere spectra with uncertainties less than 0.1% in each spectral bin. The source code is made available[4] for public scrutiny and for further development by members of the scientific community.

An improvement of the present work over previous codes is that in McPHAC, the discretization of the column depth variable in the radiative transfer problem is set independently for each discrete frequency, rather than using the same discretization as in the temperature correction procedure. Using separate discretizations for the two main components of the calculation allows the numerical uncertainty introduced in each component to be studied separately and allows independent optimization of the two components of the calculation.

A discussion of the numerical convergence of the McPHAC code is given in Section 4.3, and the results indicate that the code is capable of producing spectra with numerical uncertainties less than 0.01%. To constrain other uncertainties in the spectra, however, future work is needed. To this end, a number of possible improvements to the physical assumptions made by the code are suggested in Section 5. Several computational improvements are also suggested there, intended to reduce the computational cost of producing high-accuracy spectra on large grids of parameters.

Unlike previous unmagnetized hydrogen atmosphere models (Z96; G02; H06), the McPHAC code treats Thomson scattering anisotropically. Spectra computed assuming dipolar Thomson scattering are compared to those computed assuming isotropic scattering, and it is found that there are differences at the level of 10^{-3} below 1 keV, while the spectra vary by as much as 3% above 1 keV. For the effective temperatures for which a large fraction of the flux is expected above 1 keV ($T_{\rm eff} = 10^{6.2}$ K and $T_{\rm eff} = 10^{6.5}$ K), the differences are at the level of half a percent. It is concluded that treating Thomson scattering anisotropically is required if spectra with uncertainties less than 0.1% are to be produced, though it should be re-emphasized that there are other sources of uncertainty that also need to be reduced to produce spectra accurate to that level (see Section 1).

McPHAC-computed spectra are compared in detail to those produced in one of the previous unmagnetized hydrogen atmosphere models (Z96) in Section 4.4. It is found that the $T_{\rm eff} = 10^{5.3}$ K and $T_{\rm eff} = 10^{5.6}$ K spectra of the previous work predict significantly less flux above ~0.7 keV and ~1.4 keV, respectively, than do the spectra of the present work. At those frequencies, the spectral flux is down by a factor of ~1/30 from the peak of the spectrum, so the deficit is unlikely to be important unless only observing the tail of the spectrum. The cause of the flux deficit is determined to be that the previous work did not consider large enough column depths at those effective temperatures.

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APPENDIX A
TEMPERATURE CORRECTION PROCEDURE

A.1. Temperature Correction

The temperature correction scheme used in the present work is based on that used in previous work (Z96) and is described in detail in this section.

[4] http://dualcore.physics.mcgill.ca/McPHAC/

First, the second-order transfer equation is integrated over μ to give

$$\frac{1}{k_\nu}\frac{d}{dy}\left(\frac{1}{k_\nu}\frac{d}{dy}\int_0^1 \mu^2 P_\nu(y,\mu)d\mu\right) = J_\nu - (1-\rho_\nu)B_\nu - \frac{2\rho_\nu}{\sigma_{\mathrm{T,r}}}\int_0^1 d\mu \int_0^1 d\mu' P_\nu(y,\mu')\frac{d\sigma_{\mathrm{T,r}}}{d\mu'}(\mu,\mu'). \quad \text{(A1)}$$

The last term in the above equation must reduce to $\rho_\nu J_\nu$ since the total angle-integrated specific intensity "emitted" through scattering must equal that "absorbed." Using this fact and the definition of the Eddington factor $f_\nu = J_\nu^{-1}\int_0^1 \mu^2 P_\nu(y,\mu)d\mu$, Equation (A1) reduces to

$$\frac{1}{k_\nu}\frac{d}{dy}\left(\frac{1}{k_\nu}\frac{d}{dy}(f_\nu J_\nu)\right) = (J_\nu - B_\nu(T(y)))(1-\rho_\nu). \quad \text{(A2)}$$

To proceed, it is noted that the energy flux at a given frequency can be written as $F_\nu = (4\pi/k_\nu)(d/dy)(f_\nu J_\nu)$. Inserting this result into Equation (A2), multiplying by k_ν, and integrating over all frequencies gives

$$\frac{1}{4\pi}\frac{d}{dy}\int_0^\infty F_\nu d\nu = \int_0^\infty (J_\nu - B_\nu(T(y)))\kappa_\nu d\nu, \quad \text{(A3)}$$

since $(1-\rho_\nu)k_\nu = \kappa_\nu$. In radiative equilibrium, the energy flux through every depth of the atmosphere will be the same. Thus, the left-hand side of the above equation vanishes, so the right-hand side must also equal zero. Taking the radiative equilibrium temperature profile to be $T(y) = T_0(y) + \Delta T(y)$, where $T_0(y)$ is the present best-guess temperature profile and $\Delta T(y)$ is a correction to this profile, the BB function can be expanded to first order in a Taylor series: $B(T) = B(T_0) + dB/dT\,\Delta T$. Together with Equation (A3), this gives the expression

$$0 = \int_0^\infty (J_\nu - B_\nu(T_0) - \frac{dB_\nu}{dT}\Delta T)\kappa_\nu d\nu, \quad \text{(A4)}$$

which can be solved for the temperature correction:

$$\Delta T = \frac{\int_0^\infty (J_\nu - B_\nu(T_0))\kappa_\nu d\nu}{\int_0^\infty \frac{dB_\nu}{dT}\kappa_\nu d\nu}. \quad \text{(A5)}$$

It is important to stress that the temperature correction computed in this way will satisfy constant energy flux through the atmosphere, since this was assumed in equating the right-hand side of Equation (A3) to zero.

To find J_ν and thus ΔT, the equation to solve comes from combining Equations (A5) and (A2) with the Taylor expansion for $B(T)$ above to give

$$\frac{1}{k_\nu}\frac{d}{dy}\left(\frac{1}{k_\nu}\frac{d}{dy}(f_\nu J_\nu)\right) = \left(J_\nu - B_\nu(T_0) - \frac{dB_\nu}{dT}\frac{\int_0^\infty (J_\nu - B_\nu(T_0))\kappa_\nu d\nu}{\int_0^\infty \frac{dB_\nu}{dT}\kappa_\nu d\nu}\right)(1-\rho_\nu). \quad \text{(A6)}$$

To solve this equation, the f_ν computed by solving the full angular radiative transfer problem is used. By using a fixed f_ν, this equation can be solved using a modified (by removing dependence on an angular variable) Rybicki approach (Rybicki 1971), which will give a corrected temperature profile. The full angular radiative transfer problem can then be solved for the corrected temperature profile, which will give a new f_ν, which will allow a new temperature correction to be calculated; iteration can be continued until satisfactory convergence.

A.2. Discretization

The discretization of Equation (A6) used in McPHAC is detailed here. An alternative approach would be to use a discretization based on the Hermitian method discussed in previous work (Auer 1976), but that is not what is done in the present work.

To discretize the left-hand side of Equation (A6), $i \pm 1/2$ is used to denote evaluation at the midpoints between y_i and its nearest neighbors, such that

$$(\Delta(f_\nu J_\nu))_{i\pm 1/2} = \pm((f_\nu J_\nu)(y_{i\pm 1}) - (f_\nu J_\nu)(y_i)), \quad \text{(A7)}$$

which, along with the approximations

$$(\Delta\tau_\nu)_{i\pm 1/2} \approx k_\nu(y_{i\pm 1/2})(\Delta y)_{i\pm 1/2} \quad \text{(A8)}$$

and

$$k_{i\pm 1/2} \approx \frac{1}{2}(k_i + k_{i\pm 1}), \quad \text{(A9)}$$

gives the full discrete expression

$$\mu^2\frac{d^2}{d\tau_\nu^2}(f_\nu J_\nu) \approx \frac{\left(\frac{(f_\nu J_\nu)(y_{i+1})-(f_\nu J_\nu)(y_i)}{\frac{1}{2}[k_\nu(y_{i+1})+k_\nu(y_i)](y_{i+1}-y_i)} - \frac{(f_\nu J_\nu)(y_i)-(f_\nu J_\nu)(y_{i-1})}{\frac{1}{2}[k_\nu(y_i)+k_\nu(y_{i-1})](y_i-y_{i-1})}\right)}{\frac{1}{2}\left[\frac{1}{2}(k_\nu(y_{i+1})+k_\nu(y_i))(y_{i+1}-y_i) + \frac{1}{2}(k_\nu(y_i)+k_\nu(y_{i-1}))(y_i-y_{i-1})\right]}. \quad \text{(A10)}$$

Also approximating the integrals in Equation (A6) as discrete sums, that equation thus becomes

$$-\frac{8}{\pi\Delta_1}(f_k J_k)(y_{i-1}) + \left[\frac{1-\rho_k}{f_k(y_i)} + \frac{8}{\pi\Delta_1} + \frac{8}{\pi\Delta_2}\right](f_k J_k)(y_i) - \frac{dB_k}{dT}\frac{\Sigma_{k'=1}^K J_{k'}\kappa_{k'}b_{k'}}{\Sigma_{k'=1}^K \frac{dB_{k'}}{dT}\kappa_{k'}b_{k'}}(1-\rho_k) - \frac{8}{\pi\Delta_2}(f_k J_k)(y_{i+1}) = \left(B_k(T(y_i)) - \frac{dB_k}{dT}\frac{\Sigma_{k'=1}^K B_{k'}(T_0)\kappa_{k'}b_{k'}}{\Sigma_{k'=1}^K \frac{dB_{k'}}{dT}\kappa_{k'}b_{k'}}\right)(1-\rho_k), \quad \text{(A11)}$$

where the subscript k is used to indicate a discrete frequency point (not to be confused with the total absorption opacity k_ν), and the following definitions have been made to simplify the notation:

$$\Delta_1 = [k_\nu(y_i) + k_\nu(y_{i-1})](y_i - y_{i-1}), \quad \text{(A12)}$$

$$\Delta_2 = [k_\nu(y_{i+1}) + k_\nu(y_i)](y_{i+1} - y_i), \quad \text{(A13)}$$

$$\pi = \Delta_1 + \Delta_2. \quad \text{(A14)}$$

A.3. Rybicki Method

Proceeding as in the Feautrier method (Feautrier 1964), the above equation can be written in the form

$$\mathbf{A_i}\,\mathbf{J_{i-1}} + \mathbf{B_i}\,\mathbf{J_i} + \mathbf{C_i}\,\mathbf{J_{i+1}} = \mathbf{Q_i}, \quad \text{(A15)}$$

where $\mathbf{A_i}$, $\mathbf{B_i}$, and $\mathbf{C_i}$ are $K \times K$ matrices, $\mathbf{J_i}$ and $\mathbf{Q_i}$ are–vectors of dimension K (the number of frequency points), and $N = N_{\mathrm{temp}}$ (the number of depth points). If the boundary conditions are specified, the matrix equation can then be solved using the

Feautrier method (Feautrier 1964). However, a less computationally intensive method of solution is possible: the Rybicki method (Rybicki 1971), which is described below.

The Rybicki method (Rybicki 1971) involves restructuring the matrix problem so that most of the matrices that have to be inverted are tridiagonal matrices, which are less computationally demanding to invert than full matrices. This section describes the specific application of the previously described method (Rybicki 1971) to the temperature correction problem.

Rather than being grouped into $N_{\rm temp}$ vectors of dimension K, the values of the mean specific intensity are now written as the K column vectors

$$\mathbf{J_k} = (J_{k,1}, J_{k,2}, J_{k,3}, \ldots, J_{k,N})^T . \tag{A16}$$

In addition, one defined the vector $\bar{\mathbf{J}}$, which has elements

$$(\bar{J})_i = \frac{\sum_{k'=1}^{K} J_{k'}\kappa_{k'}b_{k'}}{\sum_{k'=1}^{K}\frac{dB_{k'}}{dT}\kappa_{k'}b_{k'}} . \tag{A17}$$

Equation (A15) can then be rewritten in the form

$$\mathbf{T_k J_k} + \mathbf{U_k}\bar{\mathbf{J}} = \mathbf{K_k}, \tag{A18}$$

where the matrix elements are given by

$$(T_k)_{i,i-1} = -\frac{8 f_k(y_{i-1}))}{\pi\Delta_1}, \tag{A19}$$

$$(T_k)_{i,i+1} = -\frac{8 f_k(y_{i+1})}{\pi\Delta_2}, \tag{A20}$$

$$(T_k)_{i,i} = \left[\frac{1-\rho_k}{f_k(y_i)} + \frac{8}{\pi\Delta_1} + \frac{8}{\pi\Delta_2}\right] f_k(y_i), \tag{A21}$$

$$(U_k)_{i,i} = -\frac{dB_k}{dT}(1-\rho_k), \tag{A22}$$

$$(K_k)_i = \left(B_k(T(y_i)) - \frac{dB_k}{dT}\frac{\sum_{k'=1}^{K} B_{k'}(T_0)\kappa_{k'}b_{k'}}{\sum_{k'=1}^{K}\frac{dB_{k'}}{dT}\kappa_{k'}b_{k'}}\right)(1-\rho_k), \tag{A23}$$

for $1 < i < N_{\rm temp}$. All other elements of $\mathbf{T_k}$ and $\mathbf{U_k}$ are zero, except those that are specified in the boundary conditions (see Section A.4).

The corresponding grand matrix equation is

$$\begin{pmatrix} \mathbf{T_1} & 0 & \ldots & 0 & \mathbf{U_1} \\ 0 & \mathbf{T_2} & \ddots & \vdots & \mathbf{U_2} \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \ldots & 0 & \mathbf{T_K} & \mathbf{U_K} \\ \mathbf{V_1} & \mathbf{V_2} & \ldots & \mathbf{V_K} & \mathbf{E} \end{pmatrix} \begin{pmatrix} \mathbf{J_1} \\ \mathbf{J_2} \\ \vdots \\ \mathbf{J_K} \\ \bar{\mathbf{J}} \end{pmatrix} = \begin{pmatrix} \mathbf{K_1} \\ \mathbf{K_2} \\ \vdots \\ \mathbf{K_K} \\ \mathbf{Y} \end{pmatrix}, \tag{A24}$$

where $\mathbf{E}$ is the negative identity matrix, $\mathbf{Y}$ is the null matrix, and

$$(V_k)_{i,i'} = \frac{\kappa_k b_k}{\sum_{k'=1}^{K}\frac{dB_{k'}}{dT}\kappa_{k'}b_{k'}}\delta_{i,i'} . \tag{A25}$$

With the addition of the boundary conditions described below, Equation (A24) can now be solved as in the standard Rybicki method (Rybicki 1971), giving $\mathbf{J_k}$, which can then be used to calculate the temperature correction.

A.4. Boundary Conditions

The boundary conditions used for the temperature correction procedure in the present work are the same as those used in a previous work (Z96), where the same temperature correction scheme was used, but with a different discretization method.

At the top of the atmosphere, the boundary condition

$$\frac{1}{k_\nu}\frac{d}{dy}(f_\nu J_\nu) = h_\nu J_\nu \tag{A26}$$

is used, while at the bottom, the condition

$$J_\nu = B_\nu \tag{A27}$$

is used.

The boundary condition at the top of the atmosphere can be discretized as

$$\frac{(f_k J_k)(y_2) - (f_k J_k)(y_1)}{\frac{1}{2}[k_k(y_2) + k_k(y_1)](y_2 - y_1)} = (h_k J_k)(y_1) . \tag{A28}$$

Thus, for first-order accurate boundary conditions, the previously unspecified $i = 1$ elements of the matrices $\mathbf{T_k}$ are specified as follows:

$$(T_k)_{1,2} = \frac{-f_k(y_2)}{\frac{1}{2}(k_\nu(y_2) + k_\nu(y_1))(y_2 - y_1)}, \tag{A29}$$

$$(T_k)_{1,1} = h_k + \frac{f_k(y_1)}{\frac{1}{2}(k_\nu(y_2) + k_\nu(y_1))(y_2 - y_1)} . \tag{A30}$$

At the bottom of the atmosphere, the boundary condition is trivially discretized, with the previously unspecified $i = N_{\rm temp}$ elements of the $\mathbf{T_K}$ matrices specified as follows:

$$(T_k)_{N,N-1} = 0, \tag{A31}$$

$$(T_k)_{N,N} = 1 . \tag{A32}$$

Finally, the $i = N_{\rm temp}$ elements of the $\mathbf{K_k}$ vectors are given by

$$(K_k)_N = B_k(T(y_N)) . \tag{A33}$$

APPENDIX B

THE CODE

This section gives a brief overview of the implementation of the McPHAC code and the numerical packages and routines that are used. The source code itself is made available online[5] and serves as its own documentation. Thus, any questions not answered in the present work should be resolvable by consulting the code.

B.1. Platform and Language

The McPHAC code is written in the C language but makes calls to several Fortran functions written by other groups. The code has been developed and tested on an AMD platform running the Linux operating system, using the GNU `gcc/gfortran` compiler to compile and link both the C and Fortran code.

[5] http://dualcore.physics.mcgill.ca/McPHAC/

B.2. External Routines

Several external routines are used for standard numerical problems, and these are described here.

B.2.1. LAPACK

The Linear Algebra PACKage (LAPACK) v3.2.1[6] routine `dgesv()` is used to invert full matrices, while the routine `dgtsv()` is used to invert tridiagonal matrices. These routines are Fortran routines and make use of several BLAS routines (see below). The source code for these routines is included with the McPHAC code, but for optimal performance precompiled optimized libraries for the appropriate platform should be obtained.

B.2.2. BLAS

The Basic Linear Algebra Subroutines (BLAS)[7] are highly optimized linear algebra routines written in Fortran. In addition to any routines called by the above-mentioned LAPACK routines, the BLAS routine `dgemm()` is used for matrix multiplication. The source codes for all the BLAS routines are included with the McPHAC code, but it is highly recommended to use precompiled optimized libraries for the appropriate platform.

B.2.3. Cubic Spline Interpolation

Cubic spline interpolation is performed using the Fortran routines `spline_cubic_set()` and `spline_cubic_val()`, which are part of the SPLINE library.[8] The codes for those two routines and any routines required by them are included with the McPHAC code (the library is licensed under the GNU LGPL license).

B.2.4. Gauss–Legendre Quadrature

To compute the abscissas and weights for the Gauss–Legendre quadrature, the Fortran routine `legendre_com()` in the QUADRULE library[9] is used. That code is licensed under the GNU LGPL license and is included with the McPHAC code.

B.2.5. Equation of State

The EOS is obtained using OPAL EOS 2001[10] (Rogers et al. 1996) routines, which are written in Fortran. The source code and associated tables are included with the McPHAC code.

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